

Analysis & Application of Learning Theory in Medical Simulation Design

Faruq Pradhan MBBCh, FRCPC

Rationale for Selection

This article (Herrera-Aliaga & Estrada, 2022) was selected given its recent publication, relevance to post-Covid 19 pandemic changes to learner needs, as well as the particularly interesting introductory hook of the evolution of simulation in education over the past century. I appreciated the mini review of the various typologies used to categorize simulations and the integration of various learning theories. The comparison of low vs medium vs high fidelity simulations was of particular interest especially on reflection of my prior personal experiences with simulation in medical education as a learner.

As an educator however, I was more particularly drawn to the zones used by the Boston Children's Simulator Program (Herrera-Aliaga & Estrada, 2022) which has helped organize my ideas for simulation education planning for my learners. Using this as a guide and keeping in mind my learners are medical students (both in pre-clinical and clinical years), I would likely be aiming for Zone 0 (basic technical skills training) and zone 1 (training of clinical competences) – especially given my early ideas on creating online asynchronous methods of simulation.

Theoretical Summary & Critique

Multiple learning theories were summarized in this paper (Herrera-Aliaga & Estrada, 2022), including those by Vytgosky, Kolb, Dreyfus and Dreyfus, Posner, Schon, Bandura, and Ericsson. For the purposes of this assignment, I will be describing the core principles of the learning theory described by Dr. K. Anders Ericsson (Ericsson, 2008). In essence, the identification and use of critical events for deliberate practice is a central concept for Ericsson's

theory on effective learning. Simulation is therefore a very effective means to be used by individuals for the deliberate practice of key tasks, actions, or skills to gain proficiency. The critical events described by Ericsson are often defined based on review of skill development of an established expert. In his 2008 paper, Ericsson provides examples of critical events in various skills (e.g., chess proficiency, typing skill, accurate musical performances) that can help measure an individual's proficiency. Though his theory is limited through its inability to explain how some people are able to achieve proficiency faster through 'intuition' or how some individuals gain technical skills with greater ease compared to others despite practicing the same tasks.

Application to Simulation Design

Combining the zones used in the Boston Children's Simulator Program (Herrera-Aliaga & Estrada, 2022) and the learning theories of Ericsson (Ericsson, 2008) there is easy applicability to medical simulation. One example for my medical students would be the use of simulation to teach bedside paracentesis (zone 0 – basic technical skills training) through deliberate practice on lifelike emulators (i.e., mannikins) with proficiency gained through repetition. This could be further developed into a clinical scenario for a patient with cirrhosis and ascites presenting with signs of sepsis with learners expected to conduct a diagnostic paracentesis and interpret the fluid studies (zone 1 – training of clinical competencies) to come to a diagnosis of spontaneous bacterial peritonitis (zone 2 – contextualized skills training in more complex/virtual environments with a focus on clinical decisions). This evolutionary simulation design can influence learner engagement given clinical applicability and allow for effective post-simulation debriefing to review key skills/concepts.

References:

Ericsson, K. A. (2008). Deliberate Practice and acquisition of Expert Performance: A General Overview. *Academic Emergency Medicine*, 15(11), 988–994.

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Herrera-Aliaga, E., & Estrada, L. D. (2022). Trends and Innovations of Simulation for twenty first century medical Education. *Frontiers in Public Health*, 10.

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